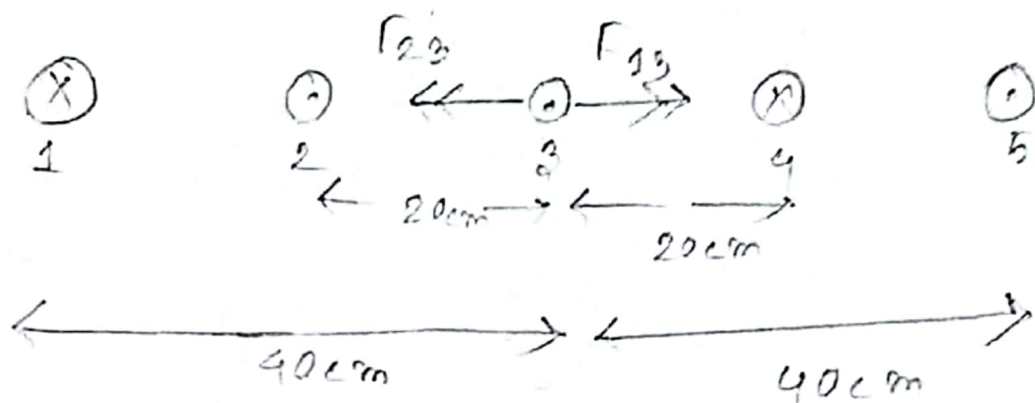


Ans. to Q. 01



$$\frac{F_{13}}{L} = \frac{\mu_0 I_1 I_3}{2\pi a_{13}} = \frac{4\pi \times 10^{-7} \times 2.5 \times 1.2}{2\pi \times 0.4} = 1.5 \times 10^{-6} \text{ Nm}^{-1} \text{ (towards right } \rightarrow \text{)}$$

$$\frac{F_{23}}{L} = \frac{\mu_0 I_2 I_3}{2\pi a_{23}} = \frac{4\pi \times 10^{-7} \times 2.5 \times 3 \times 1.2}{2\pi \times 0.2} = 3.6 \times 10^{-6} \text{ Nm}^{-1} \text{ (towards left } \leftarrow \text{)}$$

$$\frac{F_{43}}{L} = \frac{\mu_0 I_4 I_3}{2\pi a_{43}} = \frac{4\pi \times 10^{-7} \times 1.2 \times 2}{2\pi \times 0.2} = 2.4 \times 10^{-6} \text{ Nm}^{-1} \text{ (towards left } \leftarrow \text{)}$$

$$\frac{F_{53}}{L} = \frac{\mu_0 I_5 I_3}{2\pi a_{53}} = \frac{4\pi \times 10^{-7} \times 1.2 \times 2}{2\pi \times 0.4} = 1.2 \times 10^{-6} \text{ Nm}^{-1} \text{ (towards right } \rightarrow \text{)}$$

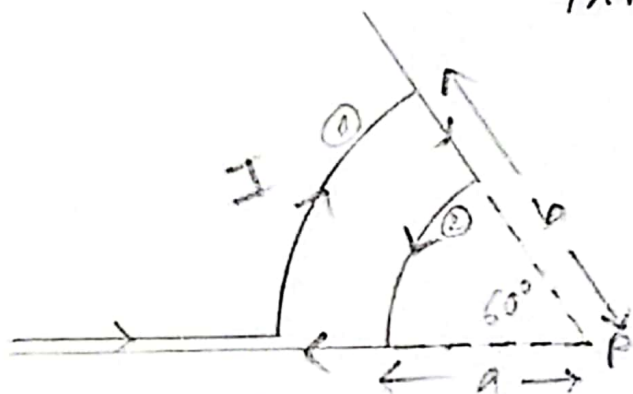
$$\begin{aligned} \therefore F_{\text{Net}} &= (F_{23} + F_{43}) - (F_{13} + F_{53}) \\ &= 6 \times 10^{-6} - 2.7 \times 10^{-6} \\ &= 3.3 \times 10^{-6} \text{ (towards left } \leftarrow \text{)} \end{aligned}$$

Ans

② Ans:

We know that,

$$B = \frac{\mu_0 I \theta}{4\pi R} \quad \theta$$



$$\begin{aligned} 1^\circ &= \frac{\pi}{180} \\ 60^\circ &= \frac{60 \times \pi}{180} \\ &= \frac{\pi}{3} \end{aligned}$$

$$B_1 = \frac{4\pi \times 10^{-7} \times 5.2}{4\pi \times 0.068} \times \frac{\pi}{3} = 8.008 \times 10^{-6} \text{ T}$$

$$B_2 = \frac{4\pi \times 10^{-7} \times 5.2}{4\pi \times 0.036} \times \frac{\pi}{3} = 1.51 \times 10^{-5} \text{ T}$$

$$\begin{aligned} \therefore B_{\text{net}} &= B_1 - B_2 \\ &= (8.008 \times 10^{-6} - 1.51 \times 10^{-5}) \text{ T} \\ &= 7.1182 \times 10^{-6} \text{ T} \end{aligned}$$

Ans

Ans. to Q. 03

① Ans:

$$\mathcal{E} = -N \frac{d\phi}{dt}$$

$$\Rightarrow IR = - \frac{d\phi}{dt}$$

$$\Rightarrow \frac{d}{dt} \cdot BA = I \frac{\mu L}{A}$$

$$\Rightarrow \frac{dB}{dt} = I \frac{\mu L}{A_w \cdot A}$$

$$\Rightarrow \frac{dB}{dt} = 1.8 \times \frac{1.72 \times 10^{-8} \times 2 (0.6 + 0.1)}{3.1416 \times (1.25 \times 10^{-3})^2 \times 0.1 \times 0.1}$$

$$= 0.3363 \text{ T/s}$$

Ans

② Ans:

